Assignment #5 for Numerical Analysis

Deadline:

1. (20 points) Solve the system by finding the LU factorization and then carrying out the two-step back substitution.

$$\begin{bmatrix} 4 & 2 & 0 \\ 4 & 4 & 2 \\ 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

2. (40 points) Consider the system

$$\begin{aligned} 0.001x_1 + 2.000x_2 + 3.000x_3 &= 1.000 \\ -1.000x_1 + 3.712x_2 + 4.623x_3 &= 2.000 \\ -2.000x_1 + 1.072x_2 + 5.643x_3 &= 3.000 \end{aligned}$$

   (a) Apply **Gaussian elimination** and **Gaussian elimination with partial pivoting** to solve the system respectively. Show your calculations and results with **four-digit accuracy**.

   (b) Use $\mathbf{e} = ||\mathbf{x} - \mathbf{x}^*||_2$ to calculate the errors, where $\mathbf{x}^* = (-0.4904, -0.05104, 0.3675)^T$.

3. (20 points) Let $\mathbf{A}$ be a symmetric matrix. After a step of Gaussian elimination, $\mathbf{A}$ reduced to

$$\begin{bmatrix} a_{11} & \mathbf{a}^T \\ \mathbf{0} & \mathbf{B} \end{bmatrix}.$$

   Prove that $\mathbf{B}$ is also a symmetric matrix.

4. (20 points) Assume that your computer can solve a 2000 x 2000 linear system Ax=b in 0.1 second. Estimate the time required to solve 100 systems of 8000 equations in 8000 unknowns with the same coefficient matrix ($Ax = b_1, ..., Ax = b_{100}$), using the LU factorization method.

5. (optional, 40 points) Suppose $\mathbf{A} \in \mathbb{C}^{m \times m}$ is strictly column diagonally dominant, which means that for each k,

$$|a_{kk}| > \sum_{j \neq k} |a_{jk}| .$$

   Show that if Gaussian elimination with partial pivoting is applied to $\mathbf{A}$, no row interchanges take place.